SACHIIN M

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Education

B.E. Computer Science & Engineering (AIML) 2023–2027
KPR Institute of Engineering and Technology, Coimbatore, India CGPA: 7.5 / 10
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Computer Networks

Work Experience

Project Intern — Payoda, Coimbatore Feb – Jul 2026
FastAPI · Node.js · Playwright · faster-whisper · LLMs

- Built a complete **e-commerce backend** using **FastAPI**, implementing structured routing, Pydantic schemas, and scalable, fault-tolerant service layers.
- Deployed a **WebSocket-based streaming architecture** for real-time speech transcription using **faster-whisper**, achieving zero API cost.
- Engineered **Node.js automation scripts** via **Playwright** for end-to-end testing across internal modules (AI TestGen, SnapTest, ScriptWeaver), and architected a **multi-agent AI system** optimizing LLM token utilization and dynamic context lengths for distributed inter-agent workflows.

Software Developer Intern — Zettaone Technologies, Bengaluru Jul – Aug 2025
Flask · React Native · OpenCV · EasyOCR · YOLOv8 · AWS S3

- Architected a **PCB component deviation detection** system using **YOLOv8 + EasyOCR**, identifying manufacturing defects in real time via a Flask-based CV pipeline in a production environment; offloaded processed results to **AWS S3** for durable storage and downstream audit.
- Engineered **VISYRE** (React Native + Flask) for QR/barcode decoding and OCR-based text extraction, eliminating manual data entry; participated in **agile sprints** with daily stand-ups and retrospectives.
- Built **Climora**, a cross-platform React Native app for real-time weather and soil insights via geolocation APIs, backed by a **microservices-style** Flask backend.

Research Intern — Providence University, Taiwan Oct – Nov 2025
CNN · PyTorch · Grad-CAM · Medical Imaging · AWS EC2

- Trained a **CNN-based pneumonia classifier** on **7,750 chest X-ray images**, achieving **92.44% overall accuracy**, **0.94 F1-score** on pneumonia class, and **0.93 weighted precision**; model training run on an **AWS EC2** GPU instance, reducing wall-clock time by 3× vs. local hardware.
- Integrated **Grad-CAM** heatmaps to localise prediction regions and generated **automated PDF medical reports** per patient for interpretable clinical diagnostic output.
- Applied SMOTE, rotation, flipping, zoom, and contrast augmentation to resolve 4:1 class imbalance; followed a **Scrum workflow** — weekly sprint planning, tasks tracked via GitHub Issues — improving team velocity by 20%.

Projects

Morpheus — Natural Language to Blender via MCP 2025
Ollama · MCP · Blender Python API (bpy) · FastAPI · React

- Integrated **Model Context Protocol (MCP)** with Blender to enable LLM-driven scene control — allowing **Ollama** (local LLM) to directly manipulate 3D objects, materials, and lighting through natural language prompts.
- Architected a **custom MCP server** bridging Ollama and Blender's Python API, enabling structured tool calls to create, modify, and render 3D scenes without manual scripting.

VoiceScript — Real-time Speech Transcription System 2025
React · FastAPI · WebSocket · faster-whisper · Silero VAD

- Built a **self-hosted real-time transcription system** using **faster-whisper** (large-v3-turbo) for accurate speech-to-text, achieving **sub-second streaming latency** with zero API cost.
- Integrated **Silero VAD** for voice activity detection and **LocalAgreement algorithm** for stable partial transcripts, eliminating word-level jitter across live audio streams.

Research Publication & Presentation

Sachiin Megarajan, Khire Rushikesh Ulhas, Ramesh Kumar T, Sutapa Debbarma, "Pneumonia Detection from Chest X-Ray Images using Dense Net," *All India Seminar on AI, Design Thinking and Innovation*, Institution of Engineers (India), Tripura State Centre Jan 2026

Skills

Languages	Python, JavaScript, Java, SQL, Node.js
Frameworks	React, React Native, FastAPI, Flask, PyTorch, TensorFlow, Angular, Playwright
Cloud / Infra	AWS (EC2, S3, Lambda, SQS, DynamoDB, IAM), Docker, GitHub Actions (CI/CD)
System Design	Microservices, REST APIs, Message Queues, Fault Tolerance, Load Balancing, Caching
ML / CV	OpenCV, YOLOv8, EasyOCR, Grad-CAM, faster-whisper, Silero VAD, Multi-Agent Systems
Tools	Git, Docker, Streamlit, VS Code, Blender (bpy), Ollama, CrewAI
Databases	MySQL, MongoDB, DynamoDB
Practices	Agile/Scrum, Sprint Planning, Code Review, GitHub Issues